

Assignment 2: Coding Basics

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OVERVIEW

This exercise accompanies the lessons/labs in Environmental Data Analytics on coding basics.

Directions

1. Rename this file `<FirstLast>_A02_CodingBasics.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.
6. After Knitting, submit the completed exercise (PDF file) to Canvas.

Basics, Part 1

1. Generate a sequence of numbers from one to 55, increasing by fives. Assign this sequence a name.
2. Compute the mean and median of this sequence.
3. Ask R to determine whether the mean is greater than the median.
4. Insert comments in your code to describe what you are doing.

```
#1.  
seqx <- seq(1,55,5) #sequence of number from 1 to 55 increasing by 5 (from,to,by);  
# call this sequence seqx  
seqx #display seqx
```

```
## [1] 1 6 11 16 21 26 31 36 41 46 51
```

```
#2.  
mean(seqx) #use function mean() to calculate the mean of seqx
```

```
## [1] 26
```

```
median(seqx) #use function median() to calculate the median of seqx
```

```
## [1] 26
```

```
#3.
mean(seqx) == median(seqx) #ask R if mean(seqx) = median (seqx) by generating true/false answer

## [1] TRUE
```

Basics, Part 2

5. Create three vectors, each with four components, consisting of (a) student names, (b) test scores, and (c) whether they are on scholarship or not (TRUE or FALSE).
6. Label each vector with a comment on what type of vector it is.
7. Combine each of the vectors into a data frame. Assign the data frame an informative name.
8. Label the columns of your data frame with informative titles.

```
student_name <- c("Isabella", "Bella", "CC", "Molly") #nominal data
test_score <- c(95, 98, 93, 90) #numerical data/ratio data
scholarship <- c(TRUE, FALSE, FALSE, FALSE) #logical data

students <- data.frame(student_name, test_score, scholarship)
names(students) <- c("Name", "Score", "On Scholarship")
students
```

```
##      Name Score On Scholarship
## 1 Isabella   95          TRUE
## 2  Bella   98          FALSE
## 3    CC    93          FALSE
## 4  Molly   90          FALSE
```

9. QUESTION: How is this data frame different from a matrix?

Answer: Data frame can hold multiple types of data by matrix can only hold one type of data. In this case, we have different vectors with multiple types of data, a data frame will be more suitable.

10. Create a function with one input. In this function, use `if...else` to evaluate the value of the input: if it is greater than 50, print the word “Pass”; otherwise print the word “Fail”.
11. Create a second function that does the exact same thing as the previous one but uses `ifelse()` instead of `if...else`.
12. Run both functions using the value 52.5 as the input
13. Run both functions using the **vector** of student test scores you created as the input. (Only one will work properly...)

```
#10. Create a function using if...else
function1 <- function(x) {
  if(x > 50) {
    print("Pass")
  }
  else {
```

```
    print("Fail")
  }
}
```

```
#11. Create a function using ifelse()
function2 <- function(x) {
  ifelse(x > 50, "Pass", "Fail")
}
```

```
#12a. Run the first function with the value 52.5
function1(52.5)
```

```
## [1] "Pass"
```

```
#12b. Run the second function with the value 52.5
function2(52.5)
```

```
## [1] "Pass"
```

```
#13a. Run the first function with the vector of test scores
# results_students <- function1(students$score)
result_Isabella <- function1(95)
```

```
## [1] "Pass"
```

```
result_Bella <- function1(98)
```

```
## [1] "Pass"
```

```
result_CC <- function1(93)
```

```
## [1] "Pass"
```

```
result_Molly <- function1(90)
```

```
## [1] "Pass"
```

```
#13b. Run the second function with the vector of test scores
results_students <- function2(students$Score)
results_students
```

```
## [1] "Pass" "Pass" "Pass" "Pass"
```

14. QUESTION: Which option of `if...else` vs. `ifelse` worked? Why? (Hint: search the web for “R vectorization”)

Answer: only ‘ifelse’ worked in terms of vectorization. ‘if...else’ can only process one numeric data at a time so it doesn’t work with vector like `students$score`.

NOTE Before knitting, you’ll need to comment out the call to the function in Q13 that does not work. (A document can’t knit if the code it contains causes an error!)